PROGRAMMING FOR DATA SCIENCE and AIML

Semester	III				
Course code					
Category	Professional Core Courses				
Course title	Programming for Data Science and AIML				
Scheme and Credits	L	T	P	Credits	
	3	0	0	3	
Classwork	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	03 Hours				

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

COURSE OBJECTIVE:

1. To impart the basic concepts of Python programming.
2. To understand concepts and usage of NumPy and Pandas packages for numerical data calculations in Python.
3. To understand concepts and applications of various data visualization tools of Python on real-world data.
4. To understand and implement the Machine Learning Concepts in Python.

UNIT - I

Overview of Python Programming Concepts: The concept of data types; variables, assignments; numerical types; operators and expressions; Control Structures; String manipulations; List; Tuple; Set Dictionaries; Functions.

UNIT - II

Introduction to Numpy: Creation on Array, Array generation from Uniform distribution, Random array generation, reshaping, maximum and minimum, reshaping, Arithmetic operations, Mathematical functions, Bracket Indexing and Selection, Broadcasting, Indexing a 2D array (matrices).

Data Manipulation with Pandas: Creating a Series - from lists, arrays and dictionaries; storing data in series from intrinsic sources, Creating Data Frames; Imputation, Grouping and aggregation, Merging, Joining, Concatenation, Find Null Values or Checking for Null Values, Reading data from CSV, txt, excel, web.

UNIT - III

Introduction to Visualization: Installing and setting up visualization libraries, Canvas and Axes, Subplots, Common plots – scatter, histogram, boxplot, Logarithmic scale, Placement of ticks and custom tick labels, Pandas Viz, Style Sheets, Plot type, Area, Bar plots, Histograms, Line Plots, Scatter Plots, Boxplots, Hexagonal Bin Plot, Kernel Density Estimation plot (KDE), Distribution

Plots, Categorical Data Plots, Combining Categorical Plots, Matrix Plots, Regression Plots, Grids; Python Visualizations toolkits/libraries.

UNIT - IV

Introduction to Machine Learning with SciKit-Learn & PyTorch: Overview of Python ML/Deep Learning toolkits/Libraries; types of machine learning algorithms: supervised, unsupervised and reinforcement learning.

Introduction to NLP with NLTK and its functions, modules like speech tagging, tokenization, parsing, segmentation, recognition, cleaning & normalization of text etc.; Overview of other Python NLP toolkits/Libraries.

COURSE OUTCOMES:

At the end of this course, students will demonstrate the ability to

CO1: Understand and implement the basics of programming in Python.
CO2: Apply the Numpy package for numerical calculations in Python.
CO3: Apply the Pandas package for loading and preprocessing data in Python.
CO4: Implement various data visualization tools of Python on real-world data.
CO5: Understand and implement the Machine Learning Concepts in Python.
CO6: Analyse day-to-day problems and industrial problems for their solutions using machine learning and data science techniques.

TEXT AND REFERENCE BOOKS:

1. Charles Dierbach, Introduction to Python using Computer Science, Wiley Publications, Second Edition, 2015
2. Mark Lutz, Learning Python, O'Reilly publications, Fifth Edition, 2015
3. Jake Vandar Plas, Python Data Science Handbook, O'Reilly, 2016
4. Paul Barry, Head First Python, O'Reilly y Publications, Second Edition, 2010

Reference Websites: (nptel, swayam, coursera, edx, udemy, official documentation weblink)

1. https://swayam.gov.in/nd1_noc19_cs59/preview
2. https://www.python.org/
3. https://www.datacamp.com/